Assignment 4 Report

Customer Churn Prediction with XGBoost

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CSCI 485 — Advanced Topics in Data Science

Part 1: Data Exploration

1.1 Dataset Overview

The IBM Telco Customer Churn dataset contains 7,043 rows and 21 columns. Each row represents one customer with demographic info, subscribed services, and billing details. The target variable is Churn (Yes/No).

1.2 Features

Feature	Type	Description
customerID	ID	Unique identifier — dropped before modeling
gender	Categorical	Male or Female
SeniorCitizen	Categorical	Whether the customer is a senior citizen (0/1)
Partner / Dependents	Categorical	Whether the customer has a partner or dependents
tenure	Numeric	Number of months the customer has stayed
PhoneService / MultipleLines	Categorical	Phone subscription details
InternetService	Categorical	DSL, Fiber optic, or No
Online Services	Categorical	OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport
StreamingTV / StreamingMovies	Categorical	Streaming subscriptions
Contract	Categorical	Month-to-month, One year, Two year
PaperlessBilling	Categorical	Whether billing is paperless
PaymentMethod	Categorical	Electronic check, Mailed check, Bank transfer, Credit card
MonthlyCharges	Numeric	Current monthly charge amount
TotalCharges	Numeric	Total charges over tenure (stored as string in raw data)
Churn	Target	Whether the customer churned — Yes=1, No=0

Table 1: All 21 features with types and descriptions.

1.3 Missing Values

After converting TotalCharges from string to numeric, 11 rows contain NaN (new customers with no charge history). All other columns have zero missing values. These 11 NaN values are handled in the preprocessing pipeline via median imputation.

1.4 Target Variable Distribution

Class	Count	Percentage
No Churn (0)	5,174	73.5%
Churn (1)	1,869	26.5%

Table 2: Churn distribution. The dataset is imbalanced — roughly 73% No, 27% Yes.

1.5 Problem Type and Business Context

This is a binary classification problem — predicting whether a customer will churn (1) or not (0). Churn prediction is critical in business because acquiring a new customer costs significantly more than retaining an existing one. By identifying at-risk customers early, the company can intervene with targeted offers or outreach to reduce revenue loss before it happens.

Part 2: Data Preprocessing

2.1 Cleaning Steps

- **Dropped customerID:** It is a unique identifier with no predictive value.
- **Encoded Churn:** Mapped Yes to 1 and No to 0 for binary classification.
- **Converted TotalCharges:** The column was stored as a string in the raw CSV. Converted to float using pd.to_numeric() with errors='coerce' so blank strings become NaN.
- **Handled NaN:** 11 missing TotalCharges values were imputed with the median inside the pipeline.

2.2 Feature Types

Numeric features: tenure, MonthlyCharges, TotalCharges

Categorical features: all remaining 16 columns including gender, Contract, InternetService, PaymentMethod, and all Yes/No service columns.

2.3 Preprocessing Pipelines

A ColumnTransformer with two sub-pipelines was built. Numeric features were imputed with the median (more robust than mean for skewed billing data). Categorical features were imputed with the most frequent value then one-hot encoded with handle_unknown='ignore' to prevent errors on unseen categories in the test set.

```
numeric_transformer = Pipeline([('imputer', SimpleImputer(strategy='median'))])
categorical_transformer = Pipeline([
    ('imputer', SimpleImputer(strategy='most_frequent')),
    ('onehot', OneHotEncoder(handle_unknown='ignore', sparse_output=False))])
```

2.4 Train/Test Split

The data was split 80/20 (train: 5,634 rows, test: 1,409 rows) using stratify=y to preserve the class imbalance ratio in both sets. random_state=42 was used for reproducibility.

Part 3: Baseline Model — Logistic Regression

Logistic Regression was trained as the baseline using max_iter=1000 to ensure convergence. It provides a linear benchmark against which XGBoost is compared.

Metric	Score
Accuracy	0.8041
Precision	0.6541
Recall	0.5561
F1-Score	0.6012
ROC-AUC	0.8426

Table 3: Logistic Regression baseline evaluation metrics.

Part 4: XGBoost Model and Hyperparameter Tuning

4.1 Hyperparameters Tuned

GridSearchCV with 3-fold cross-validation was used to tune three hyperparameters, scored by ROC-AUC to match the primary evaluation metric.

Hyperparameter	Values Tried	Justification
n_estimators	100, 200	Controls number of boosting rounds. More trees can improve fit but risk overfitting.
max_depth	3, 5	Limits tree depth to control complexity and prevent memorizing the training set.
learning_rate	0.05, 0.1	Smaller values make learning more gradual and robust, requiring more trees to compensate.

Table 4: Hyperparameters tuned with justifications.

4.2 Best Hyperparameters Found

The best combination found by GridSearchCV was: learning_rate=0.05, max_depth=3, n_estimators=100, with a cross-validated ROC-AUC of approximately 0.845. A shallower tree with a conservative learning rate performed best, suggesting the model benefits from staying simple to avoid overfitting on this dataset.

4.3 XGBoost Evaluation Results

Metric	Score
Accuracy	0.8077
Precision	0.6807
Recall	0.5187
F1-Score	0.5888
ROC-AUC	0.8458

Table 5: Tuned XGBoost evaluation metrics.

Part 5: Evaluation Metric Choice

Chosen metric: ROC-AUC

ROC-AUC is the most appropriate metric for this churn prediction problem. The dataset is class-imbalanced (73% No, 27% Yes), so accuracy alone is misleading. A naive model that always predicts No Churn would achieve 73% accuracy without identifying a single churner. ROC-AUC measures how well the model separates churners from non-churners across all classification thresholds, giving a complete picture of discriminative power. It is also useful when the business wants to rank customers by churn risk and proactively target the top at-risk segment rather than applying a fixed threshold.

Part 6: Feature Importance and Interpretation

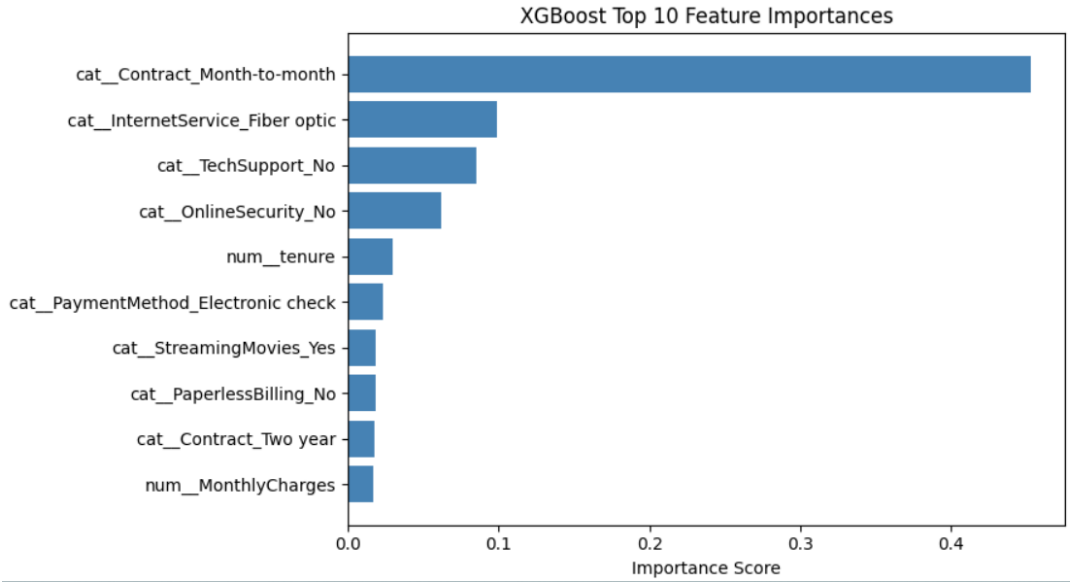
6.1 Top 10 Most Important Features

Rank	Feature	Importance Score
1	Contract — Month-to-month	0.4793
2	InternetService — Fiber optic	0.1094
3	TechSupport — No	0.0897
4	OnlineSecurity — No	0.0608
5	tenure	0.0307
6	PaymentMethod — Electronic check	0.0234
7	StreamingMovies — Yes	0.0189
8	PaperlessBilling — No	0.0182
9	Contract — Two year	0.0178
10	MonthlyCharges	0.0166

Table 6: Top 10 XGBoost feature importances by score.

6.2 Interpretation in Plain Language

- **Contract — Month-to-month (47.9%):** By far the strongest predictor. Customers on a month-to-month contract have no long-term commitment, making it effortless to leave at any time. The company should focus on converting these customers to annual plans through incentives.
- **InternetService — Fiber optic (10.9%):** Fiber optic customers churn more than DSL or no-internet customers. This likely reflects higher pricing and stronger competition in the fiber market.
- **TechSupport — No (8.9%) and OnlineSecurity — No (6.1%):** Customers without these add-on services feel less embedded in the ecosystem and perceive lower overall value. Upselling these services could improve retention.
- **tenure (3.1%):** Longer-tenured customers are less likely to churn. New customers in the first few months are the highest-risk segment and deserve early attention.
- **PaymentMethod — Electronic check (2.3%):** Customers paying by electronic check tend to churn more, possibly reflecting less engagement or financial instability.
- **Business takeaway:** The highest-risk profile is a new, month-to-month customer with fiber optic internet, paying by electronic check, and no add-on services. Retention programs should target this segment first.



Part 7: Final Comparison and Reflection

7.1 Side-by-Side Metrics

Metric	Logistic Regression	XGBoost (Tuned)	Winner
Accuracy	0.8041	0.8077	XGBoost
Precision	0.6541	0.6807	XGBoost
Recall	0.5561	0.5187	Logistic Reg.
F1-Score	0.6012	0.5888	Logistic Reg.
ROC-AUC	0.8426	0.8458	XGBoost

Table 7: Full metric comparison between Logistic Regression and XGBoost.

7.2 Which Model Performed Better?

XGBoost outperformed Logistic Regression on Accuracy, Precision, and the primary metric ROC-AUC (0.8458 vs 0.8426). Logistic Regression had slightly better Recall and F1-Score, meaning it identified more actual churners at the cost of more false positives. Overall XGBoost is the better model because ROC-AUC is the chosen primary metric and XGBoost leads on it.

7.3 Why XGBoost May Work Better on This Dataset

This dataset contains a mix of numeric and categorical features with complex non-linear interactions. For example, the combination of contract type and tenure matters more than either feature alone. Logistic Regression can only model linear decision boundaries and cannot capture these interactions without manual feature engineering. XGBoost builds an ensemble of decision trees sequentially, each correcting the errors of the previous one, naturally learning non-linear patterns and feature interactions.

7.4 One Limitation of XGBoost

XGBoost is significantly harder to interpret than Logistic Regression. While feature importances give a general sense of which variables matter, it is not straightforward to explain to a non-technical stakeholder why a specific customer was predicted to churn. In regulated industries, this lack of transparency can be a compliance barrier.